

MOHAMMED UMAIR KALEEM KHAN

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SOFTWARE ENGINEER

Authorized to work in the U.S. — no sponsorship required

PROFESSIONAL SUMMARY

Software Engineer with 2+ years building cloud-native applications, enterprise data platforms, and automation solutions using Python, TypeScript, AWS, Snowflake, dbt, SQL, and Power BI. Delivered 15+ production platform capabilities across data ingestion, observability, CI/CD, and performance engineering — collectively saving 290+ engineering hours annually, cutting incident response time by 60%, and validating platform scalability at 5,000 concurrent users.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, JavaScript, Java, C++

Cloud & Platform: AWS CDK, Bash, Lambda, S3, SQS, Linux, DynamoDB, Secrets Manager, Fargate, ECR, Infrastructure as Code (IaC), Docker.

YData & Analytics: Snowflake, Snowpark, dbt Cloud, Power BI, Pandas, ETL/ELT, data modeling, JSON, YAML

AI & Automation: Generative AI, prompt engineering, context engineering, Retrieval-Augmented Generation (RAG), Claude, GitHub Copilot, Microsoft Copilot.

Web & Frameworks: Streamlit, Vite, React, Flask, Spring Framework, REST APIs

Quality & DevOps: Pytest, Moto, Shell Scripting, Playwright, Ruff, Git, GitHub, Azure DevOps, Jenkins, CI/CD.

PROFESSIONAL EXPERIENCE

Koch Capabilities, LLC | *Software Engineer* Plano, TX | Jan 2024 – Present

- Design and deliver enterprise software, data-platform, and AI-enabled solutions using Python, TypeScript, AWS, Snowflake, Snowpark, dbt, SQL, and Power BI, supporting **14+ internal teams** across the organization.
- Designed a Streamlit roadmapping application integrated with Azure DevOps, **reducing sprint-planning effort by 48 hours annually** across two Scrum Masters and cutting backlog-visualization time from **2 hours to under 10 minutes per sprint**.
- Built a Snowpipe failure notification system with Microsoft Teams alerts and subscription workflows, **saving 120 hours annually** and reducing mean incident-detection time from **hours to under 5 minutes** across 10 monthly incidents.
- Built and maintained automation, monitoring, and operational tooling using Python, Bash, and AWS cloud services to improve platform reliability, observability, and engineering efficiency.
- Led load and performance testing with **up to 5,000 concurrent virtual users**, validating platform scalability and surfacing bottlenecks that improved sustained throughput by **35%**.
- Architected and implemented Amazon S3-to-Snowflake stage and storage integrations that enabled downstream dbt development of enterprise fact and dimension models.
- Implemented automated ingestion and conversion for structured and semi-structured files, **eliminating 10 hours of manual transformation per week** and expanding supported data workflows for data engineers.
- Developed platform integrations and automation capabilities spanning Snowflake stored procedures, Sigma APIs, and data-service workflows to improve operational efficiency and platform functionality.
- Built integration and end-to-end testing across Hub pages and platform components, **raising automated test coverage to 75%+** and catching regressions before production.
- Enhanced CI/CD pipelines using GitHub, Azure DevOps, Linux-based build environments, and automated testing to improve deployment reliability and developer productivity.
- Developed a Hub Status Page with real-time monitoring and alerting through teams' channel automatically, surfacing service health and downtime and **reducing time-to-detection for outages by 70%**.

- Created and maintained Power BI Git and deployment guidance and onboarding documentation, **reducing analyst onboarding time from 2 weeks to 3 days** across teams.

Molex LLC | *Software Engineer* May 2023 – Aug 2023

- Improved manufacturing-software productivity by **15%** through Python enhancements and SDLC optimization.
- Increased data-processing speed by **20%** using Pandas, SQL, JSON, Jenkins, Conda, Git, and Bitbucket workflows.
- Built testing and data-management solutions using Arduino and microcontroller systems, **improving validation coverage across 10+ product test cases**.
- Created tutorials and technical documentation that **increased internal tool adoption by 25%**, while contributing to code reviews and debugging.

Esprit IT Technologies | *Software Engineer* Jun 2022 – Jul 2022

- Developed manufacturing diagnostic-tool workflows with in-vehicle software specialists and supported deployment for **12 colleagues**.
- Performed regression testing and software updates using Monaco DTS8, XFlash, CANape, UDE, and Corvus across **10 McLaren Artura vehicles**.

Southern Arkansas University | *Research Assistant* Oct 2021 – May 2022

- Developed reusable Python and PyTorch components for computer-vision and deep-learning research, **reducing model prototyping time by 30%**.
- Collaborated with a 3-person team on breast-cancer detection work that achieved **89% model accuracy** and contributed to a **published paper**.

Here for the Hereafter | *ASSOCIATE DEVELOPER (VOLUNTEER)* Dec 2020 – Jan 2024

- Led the redesign and modernization of the nonprofit organization's website, improving user experience through responsive design, enhanced navigation, and integrated donation functionality.
- Coordinated Agile development activities across distributed volunteer teams, facilitating sprint planning, task prioritization, and weekly stand-ups using Asana, Slack, and Google Workspace.

SELECTED PROJECTS

Enterprise File Ingestion Platform | *Python, TypeScript, AWS S3, SQS, Lambda, Snowflake, Pytest, Playwright*

Built secure upload, processing, and raw-file retrieval workflows with presigned browser-direct uploads, multipart handling, and Excel-to-CSV conversion — supporting **5 GB uploads at 99.9% reliability** with **85%+ automated test coverage**.

Hub Observability & Status Platform | *Python, AWS, Snowflake, Microsoft Teams*

Implemented pipeline-failure notifications and service-health visibility, **saving 120 support hours annually** and cutting outage detection time by **70%**.

Roadmapper Application | *Python, Streamlit, Azure DevOps*

Designed an integrated project-planning application that **saved 48 hours annually** and reduced sprint-planning time by **90%** across two Scrum Masters.

Snowflake Platform Automation | *Python, Snowpark, Snowflake, dbt, SQL*

Created reusable utilities for table checks, auto-creation, and data-loading workflows, **reducing repetitive pipeline code by ~40%** and expanding unit-test coverage.

AMBER Alert 2.0 | *Python, Machine Learning, Google Colab*

Developed an ML-based alert-optimization prototype processing alerts in **under 2 seconds**, a **10x improvement** over the baseline workflow.

Authored and published the research paper "Modified AMBER Alerts System using TinyML Processing" in the Future Technologies Conference (FTC) 2023 Proceedings (Springer Lecture Notes in Networks and Systems).

EDUCATION

Bachelor of Science in Computer Science | Southern Arkansas University | GPA: 4.0 | Summa Cum Laude